os_lab

```
user@pc6:~$ pwd
/home/user
user@pc6:~$ cd ..
user@pc6:/home$ cd ..
user@pc6:/$ ls
bin  boot  cdrom  dev  etc  home  lib  lib64  lost+found  media  mnt  opt  pipeline  proc  root  run  sbin  snap
srv  swap.img  sys  tmp  user  usr  var
user@pc6:/$ cd home/
user@pc6:/home$ ls
user
user@pc6:/home$ cd user/
user@pc6:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  go  snap  sshkey  sshkey.pub
user@pc6:~$ cd Desktop/
user@pc6:~/Desktop$ mkdir os_lab
user@pc6:~/Desktop$ cd os_lab/
user@pc6:~/Desktop/os_lab$ mkdir -p project/src
user@pc6:~/Desktop/os_lab$ touch file1 << "hello"
>
> ^C
user@pc6:~/Desktop/os_lab$ touch file1
user@pc6:~/Desktop/os_lab$ echo "hello" >> file1
user@pc6:~/Desktop/os_lab$ cat file1
hello
user@pc6:~/Desktop/os_lab$ touch file2 file3
user@pc6:~/Desktop/os_lab$ echo "file2 here" >> file2
user@pc6:~/Desktop/os_lab$ echo "file3 here" >> file3
user@pc6:~/Desktop/os_lab$ mv file1 file2 file3 project/src
user@pc6:~/Desktop/os_lab$ ls
project
user@pc6:~/Desktop/os_lab$ chmod 700 project
user@pc6:~/Desktop/os_lab$ ls -ltrh project/
total 4.0K
drwxrwxr-x 2 user user 4.0K Oct 10 10:02 src
user@pc6:~/Desktop/os_lab$ chmod -R 1760 project
user@pc6:~/Desktop/os_lab$ ls -ltrh project/
total 4.0K
drwxrw---T 2 user user 4.0K Oct 10 10:02 src
user@pc6:~/Desktop/os_lab$ find "$HOME" -type f -user "$(whoami)" -exec chgrp adm {} \; -exec sh -c 'printf "%s\n"
 "$1" >> "$HOME/chgrp_logs.txt"' _ {} \;

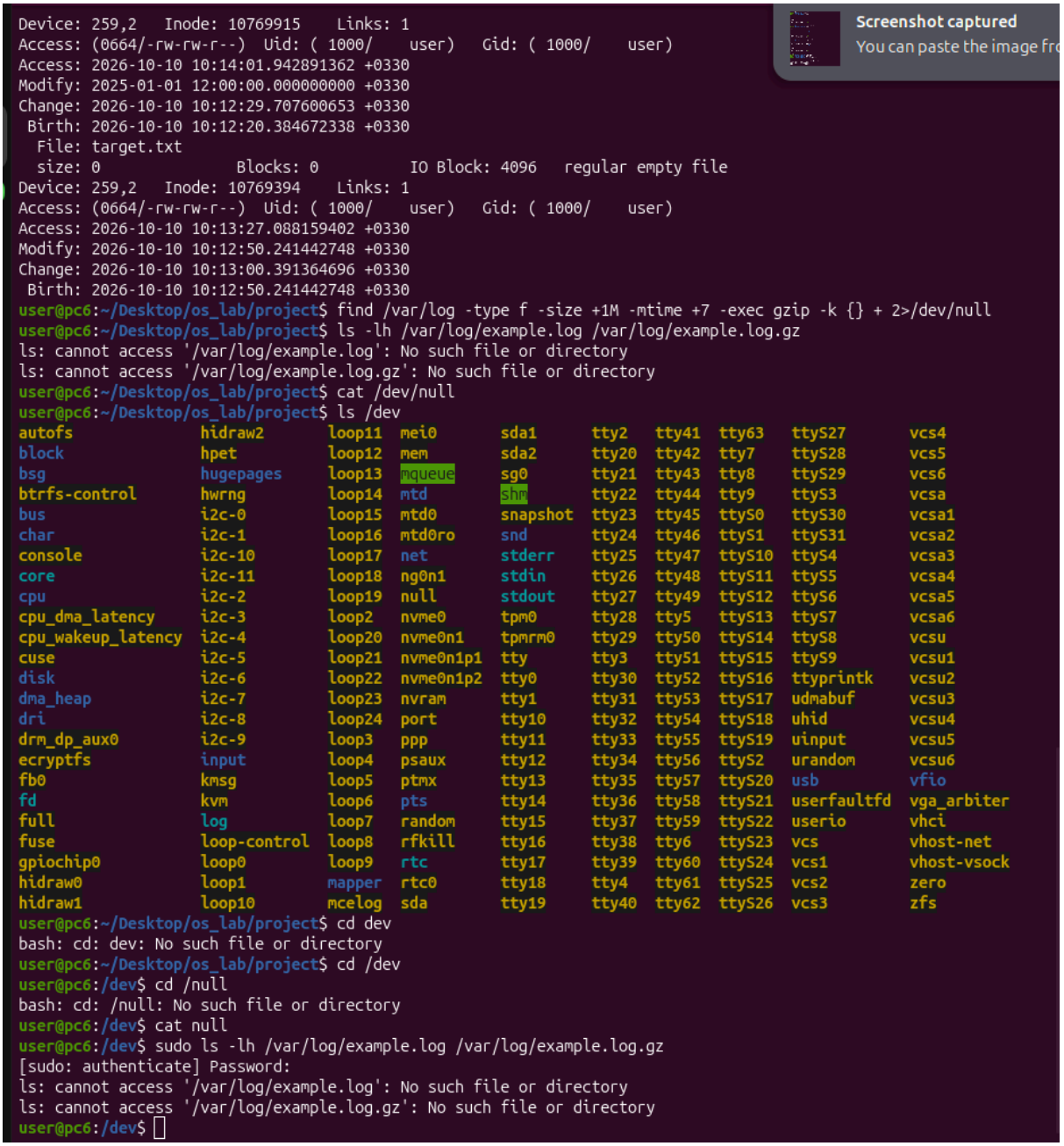

^C
user@pc6:~/Desktop/os_lab$ find "$HOME" -type f -user "$(whoami)" -exec chgrp adm {} \; -exec sh -c 'printf "%s\n"
 "$1" >> "$HOME/chgrp_logs.txt"' _ {} \;
^C
user@pc6:~/Desktop/os_lab$ find ~/Desktop/os_lab/project -type f -user "$(whoami)" -exec chgrp adm {} \; -exec sh
-c 'echo "$1" >> ~/Desktop/os_lab/project/chgrp_logs.txt' _ {} \;
user@pc6:~/Desktop/os_lab$ ls
project
user@pc6:~/Desktop/os_lab$ ls project/
chgrp_logs.txt  src
user@pc6:~/Desktop/os_lab$ cat project/chgrp_logs.txt
/home/user/Desktop/os_lab/project/src/file2
/home/user/Desktop/os_lab/project/src/file3
```

```
/home/user/Desktop/os_lab/project/src/file1
user@pc6:~/Desktop/os_lab$ cd project/
user@pc6:~/Desktop/os_lab/project$ ls -ltrhR
.:
total 8.0K
drwxrw---T 2 user user 4.0K Oct 10 10:02 src
-rw-rw-r-- 1 user user  132 Oct 10 10:10 chgrp_logs.txt

./src:
total 12K
-rwxrw---T 1 user adm  6 Oct 10 10:01 file1
-rwxrw---T 1 user adm 11 Oct 10 10:01 file2
-rwxrw---T 1 user adm 11 Oct 10 10:01 file3
user@pc6:~/Desktop/os_lab/project$ touch ref.txt
user@pc6:~/Desktop/os_lab/project$ touch -m -t "$(date -d "$(date +%Y)-01-01 -1 year" +%Y%m%d)1200" ref.txt
user@pc6:~/Desktop/os_lab/project$ ls -ltrhR
.:
total 8.0K
-rw-rw-r-- 1 user user    0 Jan  1  2025 ref.txt
drwxrw---T 2 user user 4.0K Oct 10 10:02 src
-rw-rw-r-- 1 user user  132 Oct 10 10:10 chgrp_logs.txt

./src:
total 12K
-rwxrw---T 1 user adm  6 Oct 10 10:01 file1
-rwxrw---T 1 user adm 11 Oct 10 10:01 file2
-rwxrw---T 1 user adm 11 Oct 10 10:01 file3
user@pc6:~/Desktop/os_lab/project$ touch target.txt
user@pc6:~/Desktop/os_lab/project$ touch -a -r ref.txt target.txt
user@pc6:~/Desktop/os_lab/project$ ls -ltrhR
.:
total 8.0K
-rw-rw-r-- 1 user user    0 Jan  1  2025 ref.txt
drwxrw---T 2 user user 4.0K Oct 10 10:02 src
-rw-rw-r-- 1 user user  132 Oct 10 10:10 chgrp_logs.txt
-rw-rw-r-- 1 user user    0 Oct 10 10:12 target.txt

./src:
total 12K
-rwxrw---T 1 user adm  6 Oct 10 10:01 file1
-rwxrw---T 1 user adm 11 Oct 10 10:01 file2
-rwxrw---T 1 user adm 11 Oct 10 10:01 file3
user@pc6:~/Desktop/os_lab/project$ ls target.txt
target.txt
user@pc6:~/Desktop/os_lab/project$ cat target.txt
user@pc6:~/Desktop/os_lab/project$ cat ref.txt
user@pc6:~/Desktop/os_lab/project$ stat ref.txt target.txt
  File: ref.txt
  size: 0               Blocks: 0          IO Block: 4096   regular empty file
Device: 259,2   Inode: 10769915    Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/    user)   Gid: ( 1000/    user)
Access: 2026-10-10 10:14:01.942891362 +0330
Modify: 2025-01-01 12:00:00.000000000 +0330
Change: 2026-10-10 10:12:29.707600653 +0330
 Birth: 2026-10-10 10:12:20.384672338 +0330
```



1_5:

chmod 755=> rwxr-xr-x

chmod 4755 => قبلی با بیت SUID

```
fb0                kmsg           loop5   ptmx        tty13     tty35  tty57  ttyS20  usb          vfio
fd                 kvm            loop6   pts         tty14     tty36  tty58  ttyS21  userfaultfd  vga_arbiter
full               log            loop7   random      tty15     tty37  tty59  ttyS22  userio       vhci
fuse               loop-control   loop8   rfkill      tty16     tty38  tty6   ttyS23  vcs          vhost-net
gpiochip0          loop0          loop9   rtc         tty17     tty39  tty60  ttyS24  vcs1         vhost-vsock
hidraw0            loop1          mapper  rtc0        tty18     tty4   tty61  ttyS25  vcs2         zero
hidraw1            loop10         mcelog  sda         tty19     tty40  tty62  ttyS26  vcs3         zfs
user@pc6:~/Desktop/os_lab/project$ cd dev
bash: cd: dev: No such file or directory
user@pc6:~/Desktop/os_lab/project$ cd /dev
user@pc6:/dev$ cd /null
bash: cd: /null: No such file or directory
user@pc6:/dev$ cat null
user@pc6:/dev$ sudo ls -lh /var/log/example.log /var/log/example.log.gz
[sudo: authenticate] Password:
ls: cannot access '/var/log/example.log': No such file or directory
ls: cannot access '/var/log/example.log.gz': No such file or directory
user@pc6:/dev$ find /usr/bin -type f -perm -4000 -executable 2>/dev/null > suid_binaries.txt
^C
user@pc6:/dev$ cat suid_binaries.txt
cat: suid_binaries.txt: No such file or directory
user@pc6:/dev$ sudo find /usr/bin -type f -perm -4000 -executable 2>/dev/null > suid_binaries.txt
user@pc6:/dev$ cat suid_binaries.txt
cat: suid_binaries.txt: No such file or directory
user@pc6:/dev$ printf "123\tAli\tAhmadi\t18\n124\tSara\tRezaei\t20\n125\tReza\tKarimi\t15\n126\tMina\tMoradi\t19\n
" > grade.txt
bash: grade.txt: Permission denied
user@pc6:/dev$ cd ../
user@pc6:/$ cd home/user/Desktop/os_lab/
user@pc6:~/Desktop/os_lab$ sudo find /usr/bin -type f -perm -4000 -executable 2>/dev/null > suid_binaries.txt
user@pc6:~/Desktop/os_lab$ cat suid_binaries.txt
/usr/bin/su
/usr/bin/pkexec
/usr/bin/newgidmap
/usr/bin/fusermount3
/usr/bin/umount
/usr/bin/chfn
/usr/bin/passwd
/usr/bin/sudo.ws
/usr/bin/gpasswd
/usr/bin/newgrp
/usr/bin/chsh
/usr/bin/mount
/usr/bin/newuidmap
/usr/bin/ntfs-3g
user@pc6:~/Desktop/os_lab$ printf "123\tAli\tAhmadi\t18\n124\tSara\tRezaei\t20\n125\tReza\tKarimi\t15\n126\tMina\t
Moradi\t19\n" > grade.txt
user@pc6:~/Desktop/os_lab$ cut -f1,4 grade.txt | sort -t "$(printf '\t')" -k2,2nr > top_students.txt
user@pc6:~/Desktop/os_lab$ cat top_students.txt
124     20
126     19
123     18
125     15
user@pc6:~/Desktop/os_lab$ printf "Saturday\nSunday\nMonday\nTuesday\nWednesday\nThursday\nFriday\n" > days.txt
awk -F: '$7=="/bin/bash" {print $1, $6}' /etc/passwd >> days.txt
user@pc6:~/Desktop/os_lab$ cat days.txt
```

```
user@pc6:/dev$ cd ../
user@pc6:/$ cd home/user/Desktop/os_lab/
user@pc6:~/Desktop/os_lab$ sudo find /usr/bin -type f -perm -4000 -executable 2>/dev/null > suid_binaries.txt
user@pc6:~/Desktop/os_lab$ cat suid_binaries.txt
/usr/bin/su
/usr/bin/pkexec
/usr/bin/newgidmap
/usr/bin/fusermount3
/usr/bin/umount
/usr/bin/chfn
/usr/bin/passwd
/usr/bin/sudo.ws
/usr/bin/gpasswd
/usr/bin/newgrp
/usr/bin/chsh
/usr/bin/mount
/usr/bin/newuidmap
/usr/bin/ntfs-3g
user@pc6:~/Desktop/os_lab$ printf "123\tAli\tAhmadi\t18\n124\tSara\tRezaei\t20\n125\tReza\tKarimi\t15\n126\tMina\t
Moradi\t19\n" > grade.txt
user@pc6:~/Desktop/os_lab$ cut -f1,4 grade.txt | sort -t "$(printf '\t')" -k2,2nr > top_students.txt
user@pc6:~/Desktop/os_lab$ cat top_students.txt
124     20
126     19
123     18
125     15
user@pc6:~/Desktop/os_lab$ printf "Saturday\nSunday\nMonday\nTuesday\nWednesday\nThursday\nFriday\n" > days.txt
awk -F: '$7=="/bin/bash" {print $1, $6}' /etc/passwd >> days.txt
user@pc6:~/Desktop/os_lab$ cat days.txt
Saturday
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
root /root
user /home/user
user@pc6:~/Desktop/os_lab$ find /etc -type f -name '*init*' -exec grep -Il 'systemd' {} + 2>/dev/null
/etc/systemd/system/cloud-init-network.service
user@pc6:~/Desktop/os_lab$ head -n 20 /etc/passwd | tail -n 11 | cut -d: -f1
news
uucp
proxy
www-data
backup
list
irc
_apt
nobody
systemd-network
dhcpcd
user@pc6:~/Desktop/os_lab$ tail -n 15 /etc/group | cut -d: -f1 | tr -d '\n' | wc -m
127
user@pc6:~/Desktop/os_lab$
```